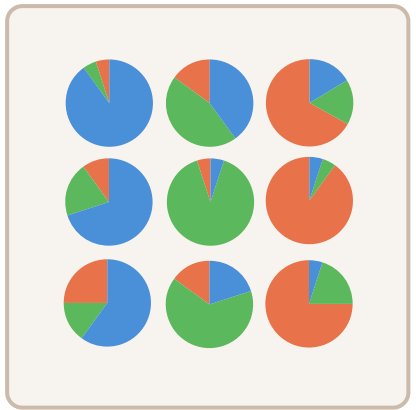
Spatial Transcriptomics

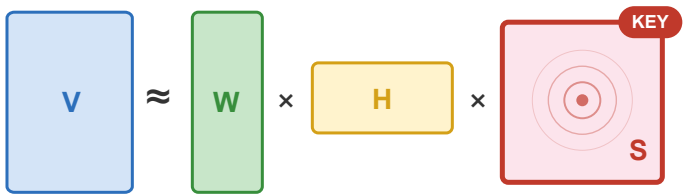
Mixed cell types per spot



SNMF

Spatial Non-negative Matrix Factorization

$$V \approx W \times H \times S$$

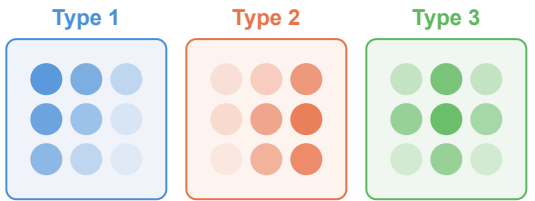


genes × spots | gene signatures | cell-type proportions | **spatial mixing matrix**

GPU-accelerated | R package | Reference-free

Minimizes KL divergence (Poisson likelihood)

Deconvolved Cell-Type Maps



Key Results

✓ **High accuracy**
Lowest RMSE on synthetic benchmarks; best ARI on real data

3× faster
GPU-accelerated; seamless CPU fallback

Biologically validated
Recovers tumor-boundary transition zone in melanoma